PYTHON FOR DATA SCIENCE · LESSON 18 OF 29

pandas III: GroupBy, Pivot Tables, Merge/Join and Reshape (melt/unpivot)

Summarise and combine data. Use groupby with agg, transform and filter, build pivot tables and crosstabs, join tables with merge (inner, left, right, outer, anti), stack with concat, and reshape between wide and long with melt, pivot, stack and unstack.

Python class 14 | pandas 2.3 / 3.0 | Intermediate | 120 min

By the end of this lesson you can

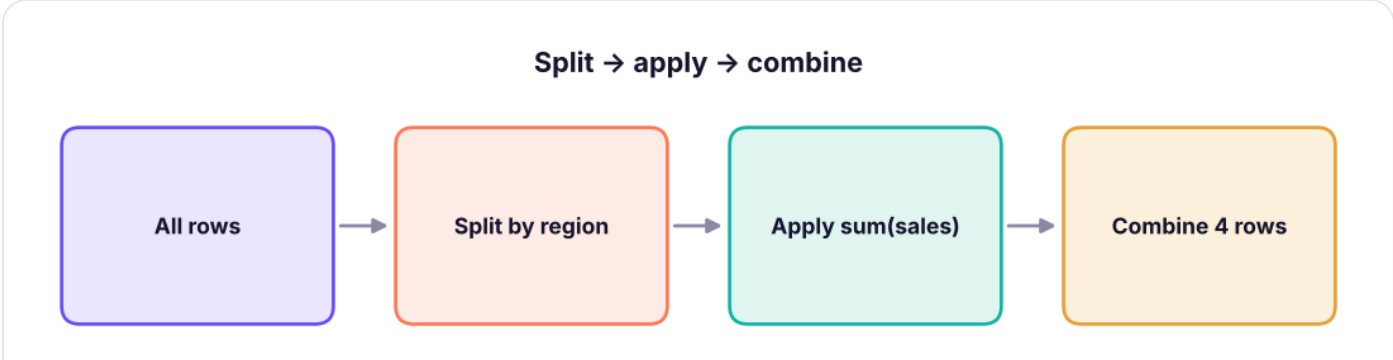
- Apply split-apply-combine with groupby, named aggregations and multiple keys
- Use transform for group-level columns and filter for group-level conditions
- Build pivot_table and crosstab summaries with totals and percentages
- Join DataFrames with merge, choose the join type, and validate the result
- Reshape wide ↔ long with melt (unpivot) and pivot, and combine with concat

Datasets used

`superstore_sales.csv` `products.csv` `customers.csv`

Split–apply–combine

`groupby` **splits** rows into groups, **applies** a calculation to each group, and **combines** the results. It is SQL's `GROUP BY` and Excel's PivotTable.



groupby splits rows by key, applies an aggregation per group, and combines the answers.

Python

```
import pandas as pd, numpy as np
pd.set_option("display.max_columns", 20); pd.set_option("display.width", 160)
s = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
print(s.groupby("region")["sales"].sum().sort_values(ascending=False))
print(s.groupby(["region", "category"])["profit"].sum().round(0).unstack())
```

OUTPUT

```
region
South    53952768.5
West     51958520.5
North    49018161.0
East     46484119.0
Name: sales, dtype: float64
category  Furniture  Office Supplies  Technology
region
East       -209426.0         434731.0   4149185.0
North      -236825.0         331848.0   4460806.0
South      -259193.0         349958.0   4611247.0
West       -112710.0         398257.0   4434286.0
```

Named aggregations

Python

```
summary = (s.groupby("category")
            .agg(orders=("order_id", "nunique"), lines=("row_id", "size"), sales=("sales",
"sum"),
                 profit=("profit", "sum"), avg_discount=("discount", "mean"), max_line=
("sales", "max"))
            .assign(margin_pct=lambda d: (100 * d.profit / d.sales).round(1))
            .sort_values("sales", ascending=False))
print(summary.round(2))
print(s.groupby("segment", as_index=False)["sales"].agg(["sum", "mean", "median",
"count"]).round(0))
```

OUTPUT

```
                orders  lines        sales       profit  avg_discount  max_line
margin_pct
category
Technology        1518   1892  156438412.5  17655523.58          0.07  619290.0
11.3
Furniture         1511   1917   34841853.0   -818153.31          0.13   93120.0
-2.3
Office Supplies   1870   2514   10133303.5   1514793.58          0.07   60830.0
```

```
14.9
       segment         sum     mean  median  count
0     Consumer  98603292.0  32066.0  8288.0   3075
1    Corporate  59113619.0  31427.0  8480.0   1881
2  Home Office  43696658.0  31965.0  9079.0   1367
```

Common aggregations: `sum`, `mean`, `median`, `min`, `max`, `count` (non-null), `size` (rows), `nunique`, `std`, `first`, `last`, or your own function.

transform and filter

`transform` returns a value **for every row** (the group result broadcast back), perfect for "share of group" and group-wise filling. `filter` keeps whole groups that meet a condition.

Python

```python
s["region_sales"] = s.groupby("region")["sales"].transform("sum")
s["share_of_region_%"] = (100 * s["sales"] / s["region_sales"]).round(3)
s["rank_in_region"] = s.groupby("region")["sales"].rank(ascending=False,
method="first").astype(int)
print(s.loc[s.rank_in_region <= 2, ["region", "product_name", "sales", "share_of_region_%",
"rank_in_region"]].sort_values(["region", "rank_in_region"]))
loyal = s.groupby("customer_id").filter(lambda g: g["order_id"].nunique() >= 8)
print(loyal["customer_id"].nunique(), "customers placed 8+ orders")
```

OUTPUT

```
     region       product_name     sales  share_of_region_%  rank_in_region
6244   East  Vertex Copier 262  472740.0              1.017               1
4219   East   Orbit Copier 746  442350.0              0.952               2
3669  North   Orbit Copier 746  619290.0              1.263               1
1653  North   Orbit Copier 746  526396.5              1.074               2
1142  South  Vertex Copier 262  551530.0              1.022               1
2087  South  Vertex Copier 262  551530.0              1.022               2
1779   West    Apex Copier 347  493080.0              0.949               1
3317   West   Orbit Copier 746  477738.0              0.919               2
41 customers placed 8+ orders
```

Pivot tables and crosstabs

Python

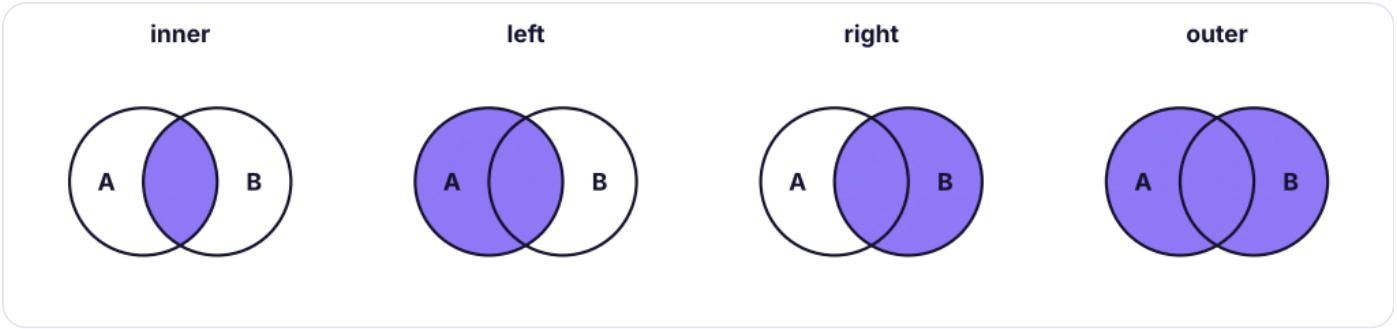
```
s["year"] = s["order_date"].dt.year
pt = pd.pivot_table(s, values="sales", index="category", columns="year", aggfunc="sum",
margins=True, margins_name="Total")
print((pt / 1e6).round(2))
print(pd.crosstab(s["segment"], s["ship_mode"], normalize="index").mul(100).round(1))
# row %
```

OUTPUT

```
year              2023   2024   2025   Total
category
Furniture         9.95  12.13  12.76   34.84
Office Supplies   2.61   3.55   3.97   10.13
Technology       45.08  49.72  61.65  156.44
Total            57.64  65.40  78.38  201.41
ship_mode    First Class  Same Day  Second Class  Standard Class
segment
Consumer            15.2       5.5          19.5            59.8
Corporate           13.2       5.0          20.3            61.5
Home Office         16.9       4.3          19.5            59.3
```

Merging (joins)

Real analysis needs several tables. `pd.merge` (or `df.merge`) joins on key columns like SQL joins.



Join types: which rows survive.

Python

```
orders = pd.DataFrame({"order_id": [1, 2, 3, 4], "cust": ["C1", "C2", "C2", "C9"],
"amount": [500, 250, 900, 120]})
cust = pd.DataFrame({"cust": ["C1", "C2", "C3"], "name": ["Asha", "Ravi", "Meera"], "city":
["Pune", "Delhi", "Kochi"]})
for how in ["inner", "left", "right", "outer"]:
    print(how, "→", len(orders.merge(cust, on="cust", how=how)), "rows")
print(orders.merge(cust, on="cust", how="left", indicator=True))
anti = orders.merge(cust, on="cust", how="left", indicator=True).query("_merge ==
'left_only'")
print("orders with unknown customer:", anti["order_id"].tolist())
```

OUTPUT

```
inner → 3 rows
left → 4 rows
right → 4 rows
outer → 5 rows
   order_id cust  amount  name   city     _merge
0         1   C1     500  Asha   Pune       both
1         2   C2     250  Ravi  Delhi       both
2         3   C2     900  Ravi  Delhi       both
3         4   C9     120   NaN    NaN  left_only
orders with unknown customer: [4]
```

Argument	Use
`on="key"` / `on=["a", "b"]`	Same column name(s) in both
`left_on=` , `right_on=`	Different key names
`how=`	`inner` , `left` , `right` , `outer` , `cross`
`suffixes=("_x", "_y")`	Clashing column names
`indicator=True`	Adds `_merge` to see where each row came from
`validate="many_to_one"`	Raises an error if the key is not unique on the right

A real three-table join

Python

```
products = pd.read_csv("products.csv"); customers = pd.read_csv("customers.csv")
lines = s[["order_id", "customer_id", "product_id", "sales", "quantity"]]
full = (lines.merge(products[["product_id", "list_price"]], on="product_id", how="left",
validate="many_to_one")
             .merge(customers[["customer_id", "city", "segment"]], on="customer_id",
how="left", validate="many_to_one"))
print(len(lines), len(full), full["list_price"].isna().sum(), "unmatched products")
full["realised_vs_list_%"] = (100 * full["sales"] / (full["list_price"] *
full["quantity"])).round(1)
print(full.groupby("segment")["realised_vs_list_%"].mean().round(1))
```

OUTPUT

```
6323 6323 0 unmatched products
segment
Consumer       91.0
Corporate      91.1
Home Office    90.4
Name: realised_vs_list_%, dtype: float64
```

WATCH OUT **Row explosion**

If the key is duplicated on **both** sides, a merge multiplies rows (many-to-many). Always compare row counts before and after, and use `validate=` to catch it.

concat: stacking tables

Python

```
jan = pd.DataFrame({"store": ["A", "B"], "sales": [100, 80]}); feb = pd.DataFrame({"store":
["A", "C"], "sales": [120, 60]})
print(pd.concat([jan.assign(month="Jan"), feb.assign(month="Feb")], ignore_index=True))
# rows (UNION ALL)
print(pd.concat([jan.set_index("store"), feb.set_index("store")], axis=1, keys=["Jan",
"Feb"]))   # columns, aligned on index
```

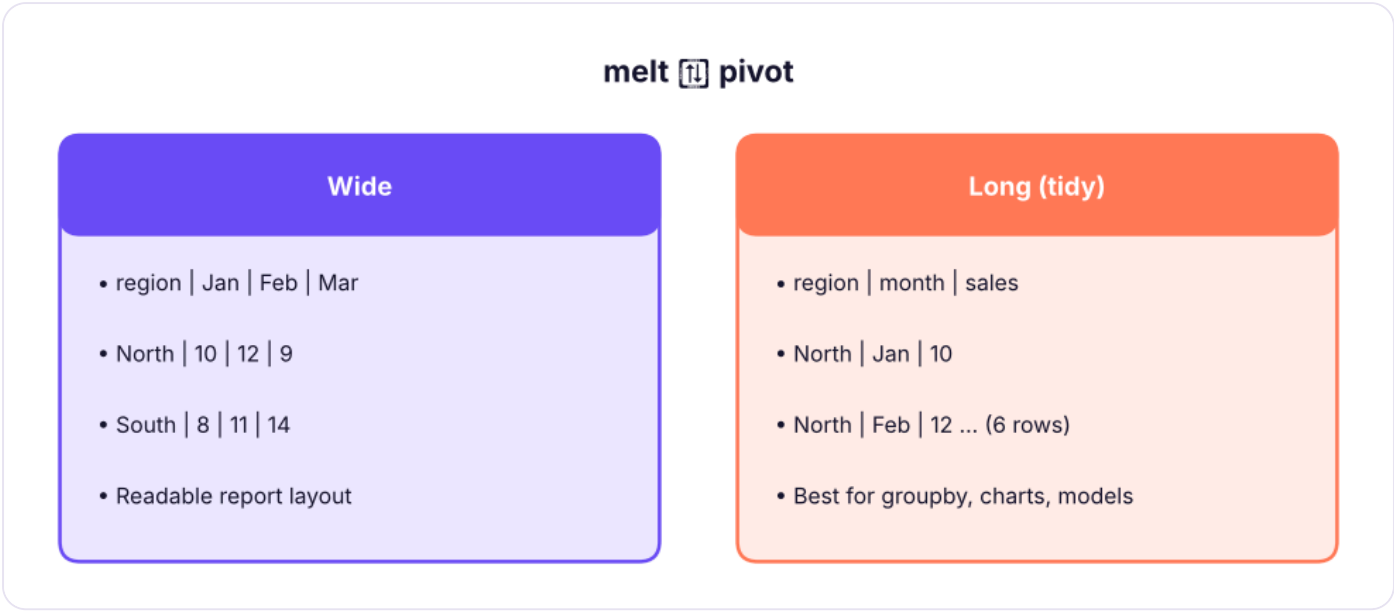
OUTPUT

```
  store  sales month
0     A    100   Jan
1     B     80   Jan
2     A    120   Feb
```

```
3     C     60   Feb
         Jan    Feb
       sales  sales
store
A      100.0  120.0
B       80.0    NaN
C        NaN   60.0
```

Reshaping: wide ↔ long

Wide tables are easy to read (one column per month); **long** (tidy) tables are easy to analyse and chart (one row per observation). `melt` is pandas' **unpivot** (the same as Power Query's "Unpivot columns").



melt turns columns into rows; pivot turns rows into columns.

Python

```python
wide = pd.DataFrame({"region": ["North", "South"], "Jan": [10, 8], "Feb": [12, 11], "Mar":
[9, 14]})
long = wide.melt(id_vars="region", var_name="month", value_name="sales")
print(long)
back = long.pivot(index="region", columns="month", values="sales")[["Jan", "Feb", "Mar"]]
print(back)
st = back.stack(); print(st.head(3)); print(st.unstack().equals(back))
```

OUTPUT

```
  region month  sales
0  North   Jan     10
1  South   Jan      8
```

```
2  North   Feb     12
3  South   Feb     11
4  North   Mar      9
5  South   Mar     14
month   Jan  Feb  Mar
region
North    10   12    9
South     8   11   14
region  month
North   Jan      10
        Feb      12
        Mar       9
dtype: int64
True
```

`pivot` requires unique index/column pairs; `pivot_table` aggregates duplicates.

Python

```python
monthly = (s.assign(month=s.order_date.dt.to_period("M").astype(str))
            .pivot_table(index="month", columns="category", values="sales",
aggfunc="sum").tail(3) / 1e6).round(2)
print(monthly)
print(monthly.reset_index().melt(id_vars="month", var_name="category",
value_name="sales_m").head(4))
```

OUTPUT

```
category  Furniture  Office Supplies  Technology
month
2025-10        1.15             0.31        4.68
2025-11        1.07             0.59        8.62
2025-12        1.31             0.40        5.94
     month         category  sales_m
0  2025-10        Furniture     1.15
1  2025-11        Furniture     1.07
2  2025-12        Furniture     1.31
3  2025-10  Office Supplies     0.31
```

Summary

KEY TAKEAWAYS

- `groupby(keys).agg(name=(col, func))` builds clean summaries; `transform` returns group values per row; `filter` keeps whole groups.
- `pivot_table` (with `margins=True`) and `crosstab` (with `normalize`) mirror Excel PivotTables.
- `merge(on=, how=)` joins tables; check row counts, use `indicator=True` for anti-joins and `validate=` to prevent row explosions.
- `concat` stacks rows (or aligns columns).
- `melt` = unpivot (wide → long); `pivot` / `pivot_table` = long → wide; `stack` / `unstack` move index levels.

Quick quiz

6 questions · tap an answer to check it

1. Which returns one value per original row?

A `groupby().agg()`

B `groupby().transform()`

C `groupby().size()`

D `pivot_table()`

Answer: B. transform broadcasts the group result back to each row.

2. Which merge keeps every order even when the customer is missing?

A inner

B left (orders on the left)

C right

D cross

Answer: B. Unmatched customer columns become NaN.

3. How do you find rows in A with no match in B?

A inner merge

B left merge with `indicator=True` and keep `_merge == "left_only"`

C outer merge without indicator

D concat

Answer: B. This is an anti-join.

4. What does `melt` do?

A Pivots long to wide

B Unpivots wide columns into rows (long format)

C Merges tables

D Deletes columns

Answer: B. `pivot` goes the other way.

5. A merge produced more rows than the left table. Most likely cause?

A A bug in pandas

B Duplicate keys on the right side

C Missing values

D Wrong dtype

Answer: B. Use `validate="many_to_one"` to catch this.

6. How do you add a grand-total row and column to a pivot table?

A `totals=True`

B `margins=True`

C `sum=True`

D `fill_value=0`

Answer: B. `margins_name` changes the label.

EXERCISE

GroupBy and merge practice

(1) Sales and profit by year and region, with margin %. (2) For each customer, the share of their spend in each category (use `transform`). (3) Top product per sub-category by profit. (4) Customers in `customers.csv` with no orders (anti-join). (5) Melt the year columns of a category × year pivot back to long format.

Show solution (worked answer)

Python

```
yr = s.groupby(["year", "region"]).agg(sales=("sales", "sum"), profit=("profit",
"sum"))
yr["margin_%"] = (100 * yr.profit / yr.sales).round(1); print(yr)
cc = s.groupby(["customer_id", "category"])["sales"].sum().reset_index()
cc["share"] = cc.sales / cc.groupby("customer_id")["sales"].transform("sum")
top = s.groupby(["sub_category", "product_name"])
["profit"].sum().reset_index().sort_values("profit").groupby("sub_category").tail(1)
no_orders = customers.merge(s[["customer_id"]].drop_duplicates(), how="left",
indicator=True).query("_merge == 'left_only'")
print(len(no_orders))
pt = s.pivot_table(index="category", columns="year", values="sales", aggfunc="sum")
print(pt.reset_index().melt(id_vars="category", var_name="year", value_name="sales"))
```

INTERVIEW CORNER

Questions you may be asked

Q: Difference between `agg` and `transform`? `agg` reduces each group to one row; `transform` returns an output aligned to the original rows.

Q: How would you compute each product's share of its category's sales? `df.sales / df.groupby("category").sales.transform("sum")`.

Q: merge vs join vs concat? `merge` joins on columns (SQL-style); `join` is a shortcut that joins on the index; `concat` stacks along an axis without matching keys.

Q: What is tidy data? Each variable is a column, each observation a row, each value a cell; long format produced by `melt` is tidy.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.